

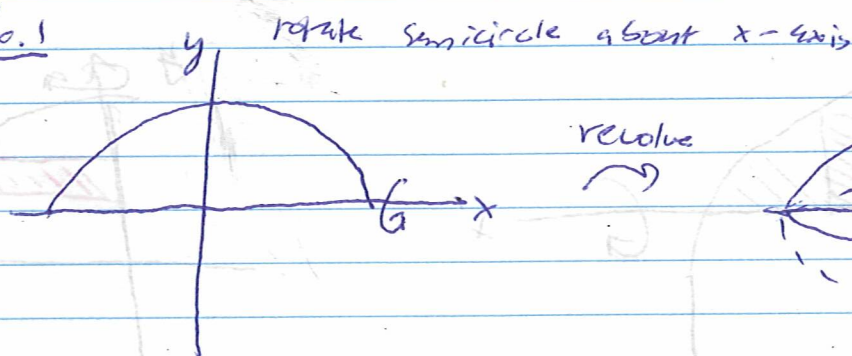
Lesson 11 - Solids of Revolution - Disk Method

10/01

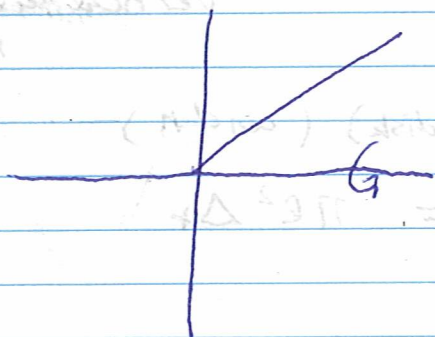
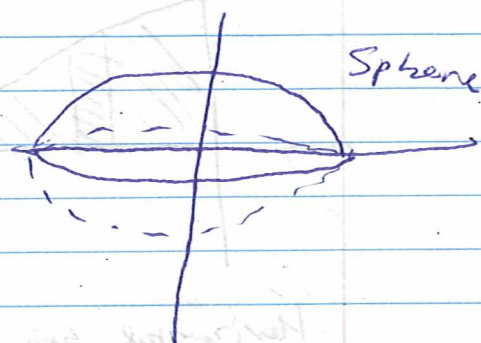
With use what we learned about area b/w. curves to find volume of solids of revolution!

If f continuous on interval $[a, b]$ If area between the graph of f and x -axis on $[a, b]$ rotated about x -axis, then a solid of revolution is generated:

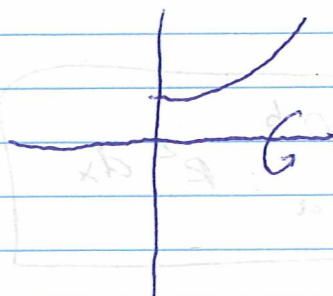
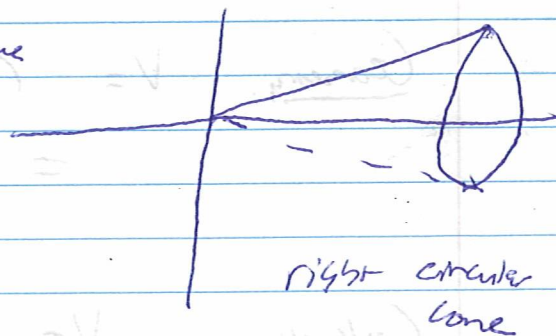
Ex. 1



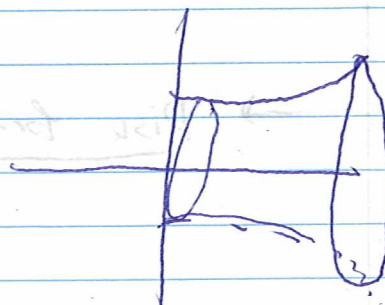
revolve
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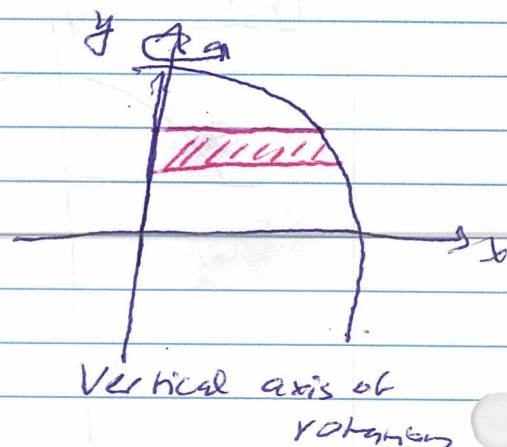
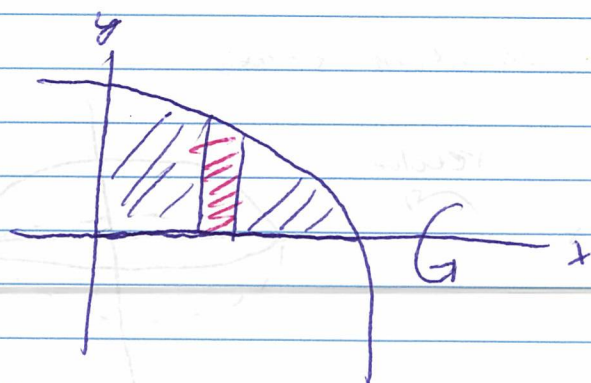
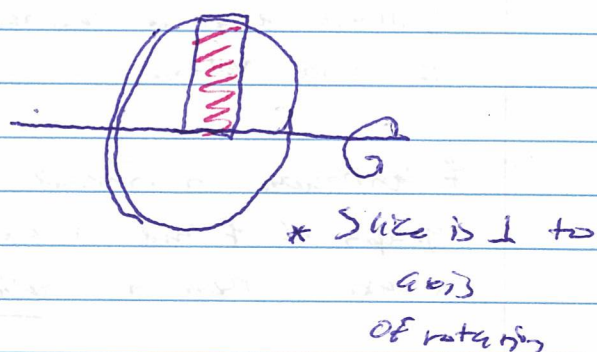
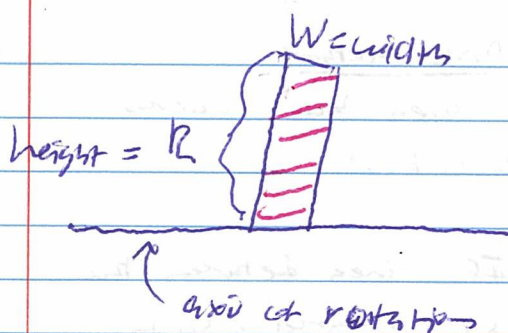
revolve
↪



revolve
↪



Goal. take a 2D region, revolve it around a horizontal or vertical line to create a 3D solid.



Geometry. $V = (\text{area of disk}) (\text{width})$
 $= \pi R^2 W = \pi R^2 \Delta x$

Calculus. $V = \lim_{n \rightarrow \infty} \sum_{i=1}^n \pi R_i^2 \Delta x_i$

$\rightarrow$ Disk formula: $V = \pi \int_a^b R^2 dx$

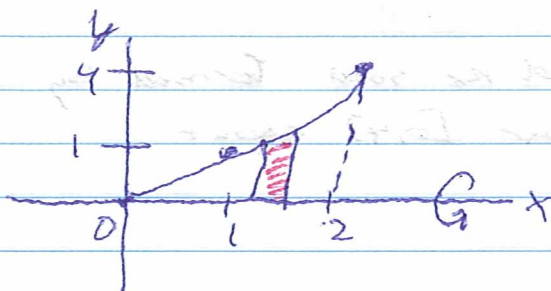
Horizontal Axis of Rotation

$$V = \pi \int_a^b R^2 dx$$

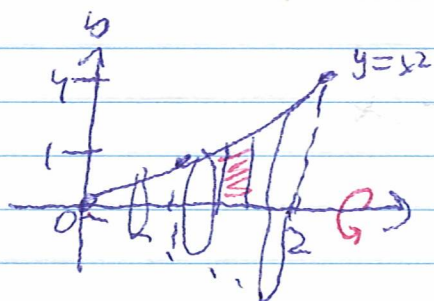
Vertical Axis of Rotation

$$V = \pi \int_c^d R^2 dy$$

Ex.1 Revolve $y=x^2$ over $[0, 2]$ about x -axis
from disk method:



x	$y=x^2$
0	0
1	1
2	4



Disk method (Horizontal axis of rotation)

$$V = \int_0^2 \pi (x^2)^2 dx$$

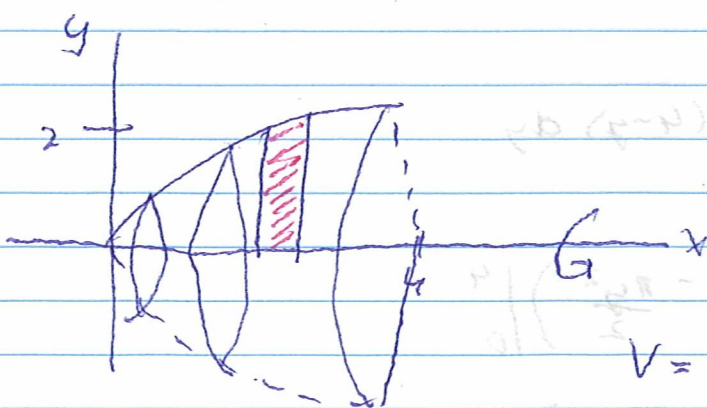
$$= \int_0^2 \pi x^4 dx$$

Radius R : $R = x^2 - 0 = x^2$

$$= \frac{\pi}{5} x^5 \Big|_0^2$$

$$= \frac{32\pi}{5} \text{ units}^3$$

Ex.2 Find volume of solid formed by
revolving $y=\sqrt{x}$ over $[1, 4]$ about x -axis.



x	$y=\sqrt{x}$
0	0
1	1
4	2

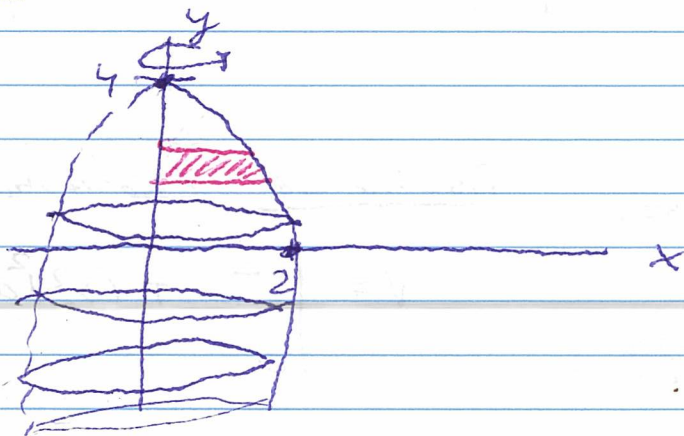
Disk method

$$V = \int_1^4 \pi (\sqrt{x})^2 dx$$

$$= \int_1^4 \pi x dx = \pi \frac{x^2}{2} \Big|_1^4$$

$$= \pi \left(\frac{16}{2} \right) - \pi \left(\frac{4}{2} \right) = \boxed{\frac{15\pi}{2}}$$

Ex-3. Find the volume of the solid formed by revolving $x = \sqrt{4-y}$ over $[0, 4]$ about y-axis.



y	$x = \sqrt{4-y}$
0	$x = \sqrt{4} = 2$ (2, 0)
4	$x = 0$ (0, 4)

Disk method : Vertical axis of rotation

$$\text{Volume} = \int_0^4 \pi (\sqrt{4-y})^2 dy$$

$$= \pi \int_0^4 (4-y) dy$$

$$= \left(4\pi y - \frac{\pi y^2}{2} \right) \Big|_0^4$$

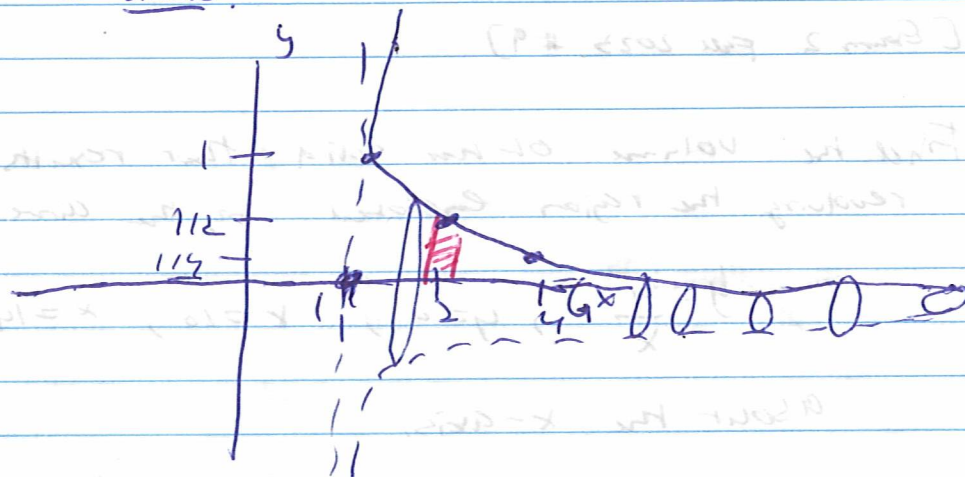
$$= 16\pi - 8\pi = \boxed{8\pi}$$

Ex. 4 [Gabriel's horn]

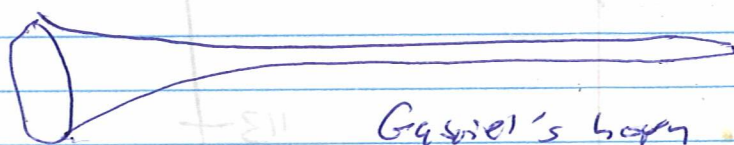
Consider the surface area and volume of the solid formed by rotating region bounded by

x -axis, $x=1$, and $y=\frac{1}{x}$ around the x -axis.

Find the volume.



x	$y = \frac{1}{x}$
1	1
2	1/2
4	1/4



Gabriel's horn

Radii: $\frac{1}{x} - 0 = \frac{1}{x}$

Disk method:

$$\text{Volume} = \int_1^{\infty} \pi \left(\frac{1}{x} \right)^2 dx$$

$$= \int_1^{\infty} \pi \frac{1}{x^2} dx$$

$$= \pi \int_1^{\infty} \frac{1}{x^2} dx$$

$$\lim_{a \rightarrow \infty} \pi \int_1^a \frac{1}{x^2} dx = \pi \left(1 - \frac{1}{a} \right) \rightarrow \boxed{\pi} \text{ as } a \rightarrow \infty.$$

Volume: π units³

Fact:

Surface area = $\boxed{\infty}$

We won't learn in this class Surface

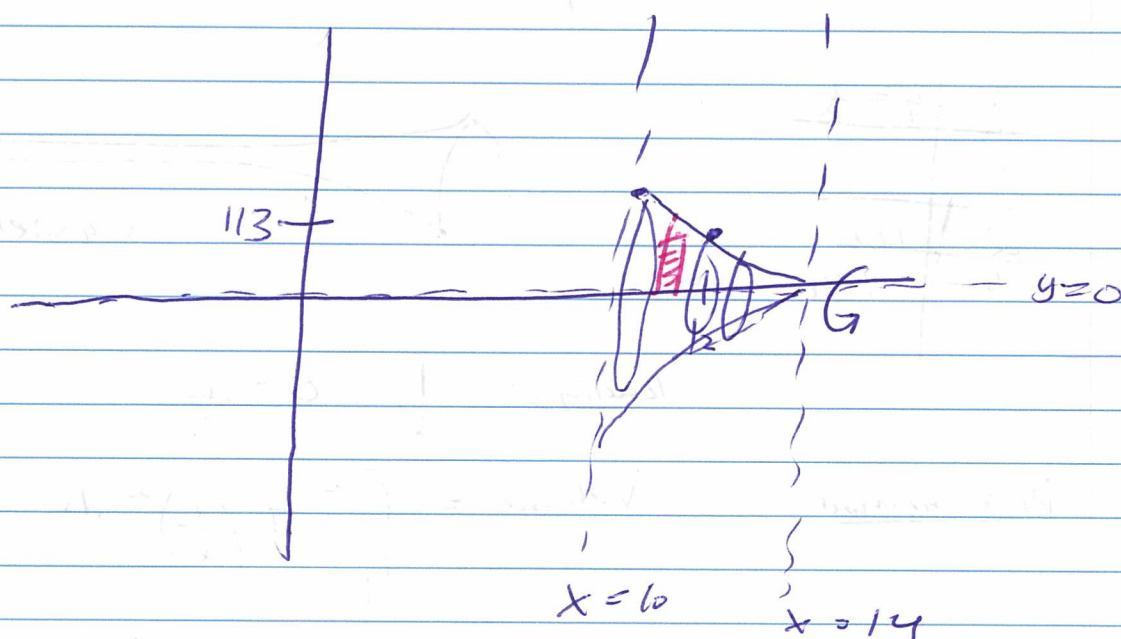
area of solids of revolution (MA 162/166
covers this)

Ex 45 [Exam 2 Fall 2023 #9]

Find the volume of the solid that results by
revolving the region enclosed by the curves

$$y = \frac{4}{x}, \quad y = 0, \quad x = 10, \quad x = 14$$

about the x -axis.



x	$y = 4/x$
12	$4/12 = 1/3$
10	$4/10 = 2/5$

Disk.

$$\begin{aligned} \text{Volume} &= \pi \int_{10}^{14} \left(\frac{4}{x} \right)^2 dx \\ &= \pi \int_{10}^{14} \frac{16}{x^2} dx \end{aligned}$$

$$= n \left[-\frac{16}{x} \right]_{10}^{14} = n \left(-\frac{16}{14} + \frac{16}{10} \right)$$

$$= 16n \left(-\frac{1}{14} + \frac{1}{10} \right)$$

$$= 16n \left(-\frac{5}{70} + \frac{7}{70} \right)$$

$$= 16n \left(\frac{2}{70} \right) = \frac{32n}{70} = \frac{16n}{35}$$

$$\boxed{\text{Volume} = \frac{16n}{35}}$$

$\boxed{\text{in exam: D}}$

1. 2011 - 2012

2. 2013 - 2014

3. 2015 - 2016

4. 2017 - 2018

5. 2019 - 2020

6. 2021 - 2022